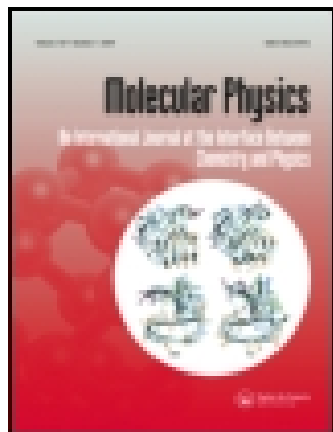




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Calculation of accurate imaginary frequencies and tunnelling coefficients for hydrogen abstraction reactions using IRCmax

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The effect of level of theory on the imaginary frequency and corresponding tunnelling coefficients has been studied for a test set of hydrogen abstraction reactions: $\bullet\text{CH}_2\text{X} + \text{CH}_3\text{Y} \rightarrow \text{CH}_3\text{X} + \bullet\text{CH}_2\text{Y}$ for $(\text{X}, \text{Y}) = (\text{H}, \text{H}), (\text{H}, \text{CN}), (\text{H}, \text{F}), (\text{H}, \text{Li})$ and (F, Li) . It is found that the imaginary frequency is very sensitive to the level of theory used, with Hartree–Fock (HF) methods severely overestimating the imaginary frequency compared with high-level CCSD(T)/6-311G(d,p) calculations. The errors for the other methods are smaller but nonetheless significant, with MP2 methods overestimating the imaginary frequency and density functional theory (DFT) methods underestimating it. In the case of the HF methods, this leads to errors in the tunnelling coefficient of several orders of magnitude, while for the better DFT and MP2 methods errors of a factor of 2–3 are observed. To address this problem, an IRCmax procedure for estimating the imaginary frequency has been developed and it is found that IRCmax imaginary frequencies calculated with CCSD(T)/6-311G(d,p) single points along a low-level HF/6-31G(d) minimum energy path provide excellent approximations to the high-level values, at a fraction of the computational cost.

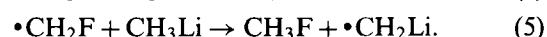
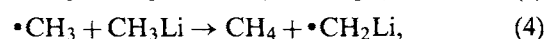
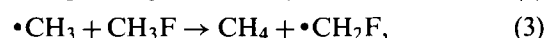
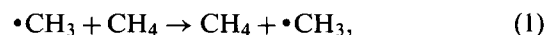
1. Introduction

Hydrogen abstraction reactions are fundamental to a variety of chemical and biochemical processes including free-radical polymerization, atmospheric chemistry, combustion chemistry and various peptide rearrangements. The calculation of accurate rate constants for these reactions provides an essential tool for probing their mechanism.

Quantum-mechanical tunnelling is an integral feature of hydrogen abstraction reactions, as the hydrogen atom is light and its wavelength is thus large compared with the barrier width (over a wide range of energies), thereby enabling it to ‘tunnel’ through the barrier. Since quantum effects in motion along the reaction coordinate are assumed to be negligible in transition state theory, the calculated rate coefficients for hydrogen abstraction reactions must be explicitly corrected for these effects via a tunnelling coefficient. There are a variety of methods for calculating the tunnelling coefficient, ranging from methods based on sophisticated multidimensional solutions to the Schrödinger equation (such as small curvature tunnelling [1], large curvature tunnelling [2] and microcanonical optimized multidimensional tunnelling [3]) to simple one-dimensional solutions using either the full minimum energy path (as in the zero-curvature

tunnelling method [4, 5]) or approximations to it (as in the Wigner [6], Bell [7] and Eckart [8] methods). The more sophisticated of these methods provide more accurate tunnelling coefficients but at the expense of requiring additional electronic structure information, such as the shape of the entire minimum energy path of the reaction rather than merely its curvature at the saddle point. Hence, the more sophisticated tunnelling methods tend to be applied at lower levels of theory. Since the accuracy of the calculated tunnelling coefficient is profoundly affected by *both* the tunnelling method used and the level of theory at which the ingredients to the calculation are obtained, choosing an accurate tunnelling method will necessarily involve a compromise between these two competing requirements.

In the present work, we examine the effect of level of theory on the accuracy of the tunnelling coefficient for a test set of hydrogen abstraction reactions:



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We focus initially on the curvature of the minimum energy path (MEP) at the saddle point of the reaction.

This quantity is widely referred to as the ‘imaginary frequency’ and is the main ingredient in all approximate one-dimensional tunnelling methods (i.e. Wigner, Bell and Eckart). The more sophisticated tunnelling methods require the shape of the entire MEP, but the imaginary frequency provides a convenient quantitative measure of this shape in the region of the transition structure where tunnelling is most significant. To show how errors in the imaginary frequency translate into errors in the tunnelling coefficients, we use the imaginary frequency to calculate Eckart tunnelling coefficients. As part of this work we show how Petersson’s IRCmax procedure [9,10] can be adapted to obtain greatly improved estimates of the imaginary frequency (and corresponding tunnelling coefficients) at a greatly reduced computational cost.

2. Computational details

Standard *ab initio* molecular orbital theory [11] and density functional theory [12] calculations were carried out using the GAUSSIAN 98 [13] and ACESII 3.0 [14] programs. Unless noted otherwise, calculations on radicals were performed with an unrestricted wavefunction. In cases where a restricted-open-shell wavefunction has been used, it is designated with an ‘R’ prefix. The frozen-core approximation was used in all calculations. Since the main objective of the present work was to assess the effect of the level of theory on the imaginary frequency of the transition structure, frequency calculations were performed at a variety of levels of theory using geometries optimized at the same level. IRCmax imaginary frequencies were also obtained via the procedure detailed below. To assist in the calculation of the tunnelling corrections, the reaction barriers and enthalpies for the various abstraction reactions were obtained with the high-level CBS-RAD procedure [15].

In performing the IRCmax procedure, MEPs were calculated using the standard GAUSSIAN ‘IRC’ keyword [16,17]. The calculations were performed for a variety of different step sizes and convergence criteria (for both the optimization convergence and the SCF convergence), in order to study the effect of these options on the accuracy of the results. The ‘IRCmax’ MEP was obtained as a series of high-level single-point energy calculations along the low-level MEP, and the imaginary frequency was estimated by fitting a parabolic function to points on this IRCmax MEP, in the vicinity of the transition structure. The model-fitting process, and our recommended optimum conditions for reproducing imaginary frequencies, will be described in the following section. All model fitting was carried out using Microsoft[®] Excel spreadsheets.

To provide an indication of how errors in the imaginary frequency translate into errors in the

tunnelling coefficient, the imaginary frequencies were converted into simple Eckart tunnelling coefficients [8]. The Eckart correction was chosen since it is generally recognized as one of the more accurate approximate one-dimensional methods. Eckart tunnelling corrections were calculated as follows. First, the imaginary frequency ($\omega^\ddagger$) is converted into the width parameter (Δ) of the Eckart function via the formula [10]:

$$\Delta = \frac{i}{2\pi c\omega^\ddagger} \left(\frac{1}{8} \frac{(B^2 - A^2)^2}{A^3} \right)^{0.5}, \quad (6)$$

where $A = (V_F^{0.5} + V_R^{0.5})^2$, $B = V_F - V_R$, V_F is the forward barrier, V_R is the reverse barrier and c is the speed of light. To simplify the later tunnelling calculations, the non-identity reactions (2)–(5) are considered in the exothermic direction, in which case $B \leq 0$. The value of Δ obtained from equation (6) is in mass-weighted units, which enables the reduced mass to be dropped from the standard Eckart formulae [8]. Hence, the permeability of the reaction barrier $G(W)$ is given as a function of particle energy W by

$$G(W) = 1 - \frac{\cosh(\alpha - \beta) + \cosh(\delta)}{\cosh(\alpha + \beta) + \cosh(\delta)}, \quad (7)$$

where $\alpha = (4\Delta\pi^2/h)(2W)^{0.5}$, $\beta = (4\Delta\pi^2/h)[2(W - B)]^{0.5}$, $\delta = (4\Delta\pi^2/h)[2A - (h^2/16\pi^2\Delta^2)]^{0.5}$ and h is Planck’s constant. The Eckart tunnelling correction (Q_t) is then obtained by numerically integrating $G(W)$ over a Boltzmann distribution of energies, via the formula [7]

$$Q_t = \frac{\exp(V_F/k_B T)}{k_B T} \int_0^\infty G(W) \exp(-W/k_B T) dW, \quad (8)$$

where k_B is the Boltzmann constant.

3. Using IRCmax to estimate imaginary frequencies

In this section, we describe how an IRCmax procedure may be used to calculate improved imaginary frequencies in a cost-effective manner. We begin by discussing the theoretical background to our procedure, which is a modification of that proposed by Petersson and co-workers [9, 10]. We then outline the mathematical procedure for estimating an imaginary frequency from a minimum energy path. Finally, we present the results of a study in which the optimum conditions for reproducing the true imaginary frequency (at a particular theoretical level) are determined.

3.1. Theoretical background

The IRCmax procedure was originally introduced to obtain improved transition structure geometries and

associated energies without the expense of a full geometry optimization [9, 10]. To achieve this, high-level single-point energy calculations are calculated not only at the transition structure (as in a traditional calculation), but also along the minimum energy path (MEP—also referred to as the intrinsic reaction coordinate or IRC). The IRCmax transition structure is then identified as the maximum of these single-point energies. This effectively optimizes the reaction coordinate—which is often the most sensitive part of a geometry optimization—at the higher level of theory, without the cost of a full optimization of all the geometric parameters at that level of theory. This approach has previously been found to provide excellent approximations to geometries that have been fully optimized at the high level of theory in a variety of different reactions, including hydrogen abstraction [9] and methyl radical addition to C=S double bonds [18].

Recently, Schwartz *et al.* [10] extended the IRCmax procedure to the prediction of simple one-dimensional Eckart tunnelling coefficients. They proposed that, rather than basing the Eckart approximation to the reaction barrier on the imaginary frequency from low-level calculations, they instead fit the Eckart function directly to an 'IRCmax' MEP—that is, to high-level single points calculated at geometries along a low-level MEP. They showed that the imaginary frequency obtained from this Eckart fit differed significantly from the corresponding low-level imaginary frequency, leading to changes in the calculated tunnelling coefficients of more than an order of magnitude.

There are two separate issues involved in estimating the tunnelling coefficient via an Eckart fit to the MEP. The first is whether to bias the fit of the Eckart function towards the points on the MEP that are nearest the transition structure (where tunnelling is most significant), or to fit a much broader portion of the MEP (but at the cost of having a poorer fit to the most significant section, near the TS). The decision on this issue can alter the resulting tunnelling coefficient by several orders of magnitude in hydrogen abstraction reactions such as those considered in the present work. This problem, which highlights the deficiency of the Eckart approach to calculating tunnelling coefficients, will be discussed in a forthcoming assessment of the effect of tunnelling method on the accuracy of tunnelling coefficients [19]. In the present work, we are instead concerned with the second issue: the accuracy of the MEP used to estimate the tunnelling coefficient. To quantify this, we compare the imaginary frequencies calculated at various low levels of theory (including 'IRCmax' imaginary frequencies) to that obtained from a full frequency calculation at a high level of theory. The procedure in the present work was thus developed with a view to obtaining

'IRCmax' imaginary frequencies that provide a consistent and objective measure of the curvature of the MEP at the transition structure.

3.2. Mathematical procedure

The MEP in the form of energy (V , in hartrees) versus intrinsic reaction coordinate (s , in bohr amu^{0.5}) is fitted with a parabolic function (of the form $V(s) = as^2 + bs + c$). The parabolic function was chosen over the Eckart function since, under the harmonic oscillator approximation, this is the functional form of the vibrational motion. Since the MEP is already in mass-weighted units, the second derivative of $V(s)$, $2a$, provides a measure of the mass-weighted force constant $F_\mu^\ddagger$. The imaginary frequency ($\omega^\ddagger$) is then simply given as

$$\omega^\ddagger = \frac{1}{2\pi c} (F_\mu^\ddagger)^{0.5} \quad (9)$$

(where c is the speed of light). To obtain $\omega^\ddagger$ in cm⁻¹, the value of a obtained from the parabolic fit is multiplied by a conversion factor of 1 hartree bohr⁻² amu⁻¹ = 9.3758 × 10²⁹ s⁻².

3.3. Optimization of the technique

The extent to which the estimated imaginary frequency reflects the curvature of the MEP at the transition structure depends on which portion of the MEP is included in the model fitting. To determine the appropriate portion for approximating most closely the true imaginary frequency, we attempted to reproduce the exact imaginary frequency at a specific level of theory for a test reaction by fitting a parabola to the MEP of that reaction (as calculated at the *same* level of theory). We also used the test reaction to study the effect of other variables such as the step size and convergence criteria used in calculating the MEP. The test reaction chosen was reaction (2) ($\bullet\text{CH}_3 + \text{CH}_3\text{CN} \rightarrow \text{CH}_4 + \bullet\text{CH}_2\text{CN}$), which has an asymmetric MEP and is thus likely to provide a critical test of the model-fitting procedure.

Table 1 shows the imaginary frequency for the test reaction (at three different levels of theory), as estimated by fitting the parabolic function to increasingly larger portions of the MEP. Also included in the table are the corresponding exact imaginary frequencies, as obtained from a full frequency calculation at the same level of theory. Since it has been noted that the accuracy of the MEP is greatly affected by the step size and convergence criteria in both the original TS optimization and in the optimization steps in the IRC calculation [20], all MEPs were calculated with the smallest possible step size (0.01 bohr amu^{0.5}) and the tightest possible convergence criteria.

Table 1. Effect of the portion of the MEP included in the model fitting on the accuracy of the estimated imaginary frequency (cm^{-1}) for the hydrogen abstraction reaction $\bullet\text{CH}_3 + \text{CH}_3\text{CN} \rightarrow \text{CH}_4 + \bullet\text{CH}_2\text{CN}$.

Portion of MEP fitted ^a	Level of theory used to calculate the MEP ^b		
	HF/ 6-31G(d)	B3-LYP/ 6-31G(2df,p)	MP2/ 6-311G(d,p)
$s = -0.01$ to $+0.01$	1827i	735i	1606i
$s = -0.05$ to $+0.05$	2422i	1504i	1806i
$s = -0.10$ to $+0.10$	2475i	1523i	1809i
$s = -0.25$ to $+0.25$	2373i	1441i	1668i
$s = -0.50$ to $+0.50$	1927i	1194i	1283i
Exact imaginary frequency ^c	2578i	1568i	1866i

^a Given as the range of values of the intrinsic reaction coordinate (s in bohr $\text{amu}^{0.5}$) about the transition structure ($s = 0$).

^b All MEPs were calculated with a step size of 0.01 bohr $\text{amu}^{0.5}$, using the 'verytight' convergence criteria for both the SCF and geometry optimization step of the MEP calculation.

^c This is the analytically calculated imaginary frequency at the particular level of theory.

From an examination of these results, it can be seen that imaginary frequencies estimated from the smallest possible portion of the MEP (i.e. the transition structure and the points at just one step of 0.01 bohr $\text{amu}^{0.5}$ to either side of it) greatly underestimate the imaginary frequency—by as much as 50% in the case of the B3-LYP/6-31G(2df,p) surface. As the portion of the MEP is increased, the estimated imaginary frequency increases to a maximum before, on further increasing the range of values fitted, decreasing again. The maximum values of the imaginary frequency occur for fitting to ± 0.11 bohr $\text{amu}^{0.5}$ for the HF/6-31G(d) surface, ± 0.09 bohr $\text{amu}^{0.5}$ for the B3-LYP/6-31G(2df,p) surface and ± 0.07 bohr $\text{amu}^{0.5}$ for the MP2/6-311G(d,p) surface, and are 2476i, 1523i and 1814i cm^{-1} , respectively. These optimum values underestimate the corresponding exact value for the imaginary frequency at the particular levels of theory by around 3% in all cases.

The results above indicate that the optimum range for fitting to the MEP is around ± 0.10 bohr $\text{amu}^{0.5}$, with large errors occurring if the ranges are much shorter or much longer. Apparently, the IRC algorithm [16,17] is not very accurate for the first few steps away from the saddle point. Examination of the resulting molecular configurations shows that the path length, s , does not correspond to the nominal value initially. However, the algorithm subsequently converges to the correct path, so that by the time the coordinate s reaches ± 0.10 bohr $\text{amu}^{0.5}$, a reasonable approximation to the correct path is followed. However, beyond this point, a competing effect starts to take over. The imaginary

frequency is defined as the curvature of the MEP at the transition structure. Since the MEP is not parabolic over larger distances, the fitting of a parabolic function to points over the broader MEP provides an increasingly crude approximation to the curvature at the transition structure. The best results therefore occur at intermediate displacements and are necessarily a compromise between these two competing effects—which may be the cause of the residual error of 3%. As will be seen later (and as is also apparent from the variation in the exact frequency with level of theory, shown in table 1), this residual error in estimating an imaginary frequency is much smaller than the errors due to using a poor level of theory and thus do not significantly compromise the accuracy of the IRCmax technique.

Thus far, imaginary frequencies have been estimated from MEPs calculated with the smallest possible step size and tightest possible convergence criteria. As noted above, these were selected because they have previously been shown to be important in obtaining an accurate description of the reaction path [20]. However, since these add significantly to the cost of the calculation, it is of interest to examine whether or not looser convergence criteria and/or larger step sizes are acceptable. Table 2 shows the estimated imaginary frequencies for the test reaction at the B3-LYP/6-31G(2df,p) level of theory, as obtained from MEPs calculated with a variety of step sizes. The exact imaginary frequency at this level is included in the table for purposes of comparison. In all cases, the portion of the MEP to be fitted was selected so as to provide the most accurate estimate of the imaginary frequency. From these results, it can be seen that the error in the estimated imaginary frequency doubles upon increasing the step size from 0.01 to 0.02 bohr $\text{amu}^{0.5}$, and doubles again on increasing the

Table 2. Effect of the IRC step size on the accuracy of the estimated imaginary frequency ($\omega^\ddagger$, cm^{-1})^a for the hydrogen abstraction reaction $\bullet\text{CH}_3 + \text{CH}_3\text{CN} \rightarrow \text{CH}_4 + \bullet\text{CH}_2\text{CN}$.

Step size	$\omega^\ddagger$	% Error in $\omega^\ddagger$	Portion of MEP fitted ^b
0.01 bohr $\text{amu}^{0.5}$	1523i	3	± 0.10 bohr $\text{amu}^{0.5}$
0.02 bohr $\text{amu}^{0.5}$	1468i	6	± 0.16 bohr $\text{amu}^{0.5}$
0.05 bohr $\text{amu}^{0.5}$	1397i	11	± 0.20 bohr $\text{amu}^{0.5}$
0.10 bohr $\text{amu}^{0.5}$	1398i	11	± 0.20 bohr $\text{amu}^{0.5}$
Exact imaginary frequency	1568i	—	—

^a Calculated by fitting a parabolic function to various B3-LYP/6-31G(2df,p) MEPs, obtained with various step sizes (as shown) and using the tightest possible convergence criteria for both the SCF and optimization convergence in all cases.

^b The portion of the MEP fitted (given as a range of s values to either side of the transition structure) was chosen so as to give the best possible value of $\omega^\ddagger$ (compared with the exact imaginary frequency) in each case.

step size to 0.05 or 0.10 bohr amu^{0.5}. The larger step sizes introduce further error into the results because they exacerbate the problems caused by the approximate nature of the initial step in the MEP algorithm, thereby causing the MEP to take longer to return to the correct reaction path. This view is consistent with the fact that the optimum range of points for obtaining the best imaginary frequency is increased for the larger step sizes.

Limited testing was also performed to examine the effect of the convergence criteria (for both the SCF and the optimization steps along the MEP) on the accuracy of the estimated imaginary frequencies. Using the smallest possible step size and fitting to the optimum portion of the MEP, the imaginary frequency was estimated from HF/6-31G(d) MEPs that were calculated using a variety of different convergence criteria. It was found that, at least for the test reaction, the estimated imaginary frequency was insensitive to the SCF convergence criteria, with the results using the default criteria and the tightest sensible criteria ('Conver = 12') agreeing to within 0.01%. Likewise, the results obtained using the default convergence criteria and tightest sensible criteria ('verytight') for the optimization steps along the MEP agreed to within 0.05%. Hence, at least for the test reaction, it should be possible to use the default convergence criteria without introducing significant errors to the results.

In summary, it is possible to estimate an imaginary frequency by fitting a parabolic function to a MEP. For the best results, the MEP should be calculated using the smallest possible step size (0.01 bohr amu^{0.5}) and the parabola fitted to points within ± 0.10 bohr amu^{0.5} of the transition structure. Under these conditions, the resulting imaginary frequency might be expected to

underestimate the true imaginary frequency by around 3%. The present results suggest that the default optimization and SCF convergence criteria should be sufficient when calculating the MEPs. However, further testing is required to establish whether this conclusion may be generalized to other reactions and, in the present work, the tightest possible convergence criteria have in fact been used for all reactions.

4. Effect of level of theory on the imaginary frequency

In this section, the effect of the level of theory on the accuracy of the calculated imaginary frequency is examined for the hydrogen abstraction reactions (1)–(5). The geometries of the transition structures for these reactions are displayed schematically in figure 1. It should be noted that, for the sake of consistency, the C_s²A' structure for **2** was used at all levels of theory, though it is only the preferred geometry at the lower (i.e. HF, MP2 and various DFT) levels. Barriers and enthalpies for these reactions, as calculated at the CBS-

Table 3. Reaction barriers and enthalpies^a (0 K, kJ mol⁻¹) for the hydrogen abstraction reactions.

Reaction	Fwd. barrier	Rev. barrier	Enthalpy
(1) •CH ₃ + CH ₄ → CH ₄ + •CH ₃	71.7	71.7	0.0
(2) •CH ₃ + CH ₃ CN → CH ₄ + •CH ₂ CN	49.7	86.8	-37.1
(3) •CH ₃ + CH ₃ F → CH ₄ + •CH ₂ F	62.0	76.0	-14.0
(4) •CH ₃ + CH ₃ Li → CH ₄ + •CH ₂ Li	44.5	78.2	-33.6
(5) •CH ₂ F + CH ₃ Li → CH ₃ F + •CH ₂ Li	6.8	26.4	-19.7

^a Calculated using the CBS-RAD procedure and including zero-point vibrational energy corrections calculated using B3-LYP/6-31G(d) frequencies scaled by 0.9806 [21].

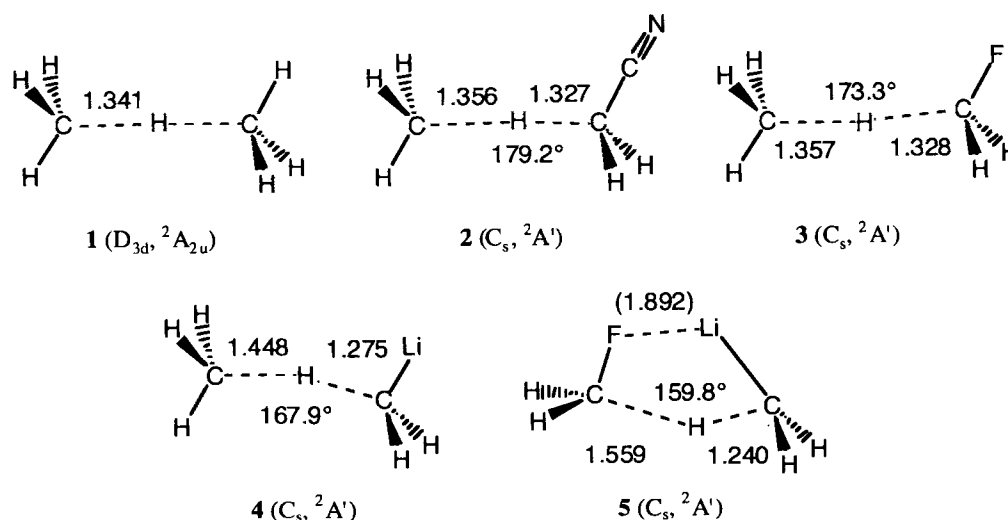


Figure 1. Optimized CCSD(T)/6-311G(d,p) transition structures for the hydrogen abstraction reactions (all bond lengths in Å).

RAD level of theory, are shown in table 3. These will be discussed in more detail in a forthcoming publication [22], but are included in the present work so that Eckart tunnelling coefficients can be calculated. However, it may be observed from table 3 that the current test set spans a range of symmetric, approximately symmetric and heavily asymmetric reaction paths, and also spans a wide range of reaction barriers (from 86.8–6.8 kJ mol⁻¹). Hence the results from this assessment should hopefully be applicable to the more general problem of modelling hydrogen-atom transfer between carbon-centred radicals.

4.1. Imaginary frequencies from full frequency calculations

The imaginary frequencies for reactions (1)–(5), as calculated at a variety of levels of theory, are shown in table 4. Results at the highest level of theory, namely CCSD(T)/6-311G(d,p), are treated as reference values. It can be seen from table 4 that the imaginary frequency is very sensitive to the level of theory, the calculated values spanning a very wide range for each reaction. In general, the RHF and UHF methods severely overestimate the imaginary frequency, the MP2 methods also overestimate it (although not as severely as the HF methods), and the various DFT methods significantly underestimate it. Of the various methods examined, the DFT procedure MPW1K [23] appears to provide the

best approximation to CCSD(T)/6-311G(d,p), producing imaginary frequencies that are usually only 5–7% too low. However, this method (and indeed the other DFT method tested, B3-LYP, with most basis sets) performs poorly for the strongly polar $\bullet\text{CH}_2\text{F} + \text{CH}_3\text{Li}$ reaction (5) where the frequency is underestimated by more than 20%. This exacerbated flattening of the DFT-calculated reaction path for reactions displaying strong polar interactions has been observed previously in studies of hydrogen abstraction barriers [24]. The MP2 methods with either moderate or large basis sets also produce reasonable absolute values for the imaginary frequency, although the errors are more variable (ranging from zero to 17%).

In general, the low-level estimates of the imaginary frequency show deviations from the corresponding high-level CCSD(T)/6-311G(d,p) values which are not consistent. The scale factors required to convert each low-level imaginary frequency into the corresponding CCSD(T)/6-311G(d,p) values are shown in parentheses in table 4. It can be seen that the scale factors at a specific level of theory generally vary considerably for these five closely related reactions and hence the outlook for the development of more general scale factors for the imaginary frequencies at most levels of theory does not look particularly promising. This is in stark contrast to the case of real frequencies (and related quantities such as zero-point vibrational energies), for which scale

Table 4. Effect of level of theory on the imaginary frequency (cm⁻¹) for the hydrogen abstraction reactions $\bullet\text{CH}_2\text{X} + \text{CH}_3\text{Y} \rightarrow \text{CH}_3\text{X} + \bullet\text{CH}_2\text{Y}$.^a

Level of theory	Reaction: $\bullet\text{CH}_2\text{X} + \text{CH}_3\text{Y} \rightarrow \text{CH}_3\text{X} + \bullet\text{CH}_2\text{Y}$				
	X,Y = H,H	X,Y = H,CN	X,Y = H,F	X,Y = H,Li	X,Y = F,Li
	(1)	(2)	(3)	(4)	(5)
RHF/6-31G(d)	3489i (0.53)	3376i (0.55)	3592i (0.54)	3338i (0.47)	3418i (0.31)
RHF/6-311G(d,p)	3411i (0.55)	3315i (0.56)	3529i (0.55)	3274i (0.48)	3376i (0.31)
RHF/6-311 + G(3df,2pd)	3398i (0.55)	3309i (0.56)	3528i (0.55)	3257i (0.48)	3389i (0.31)
UHF/6-31G(d)	2558i (0.73)	2578i (0.72)	2611i (0.74)	2421i (0.65)	2096i (0.50)
UHF/6-311G(d,p)	2518i (0.74)	2537i (0.73)	2572i (0.75)	2382i (0.66)	2070i (0.51)
UHF/6-311 + G(3df,2pd)	2530i (0.74)	2545i (0.73)	2585i (0.74)	2397i (0.66)	2110i (0.50)
RB3-LYP/6-31G(d)	1741i (1.07)	1658i (1.12)	1760i (1.09)	1368i (1.15)	796i (1.31)
RB3-LYP/6-311G(d,p)	1713i (1.09)	1640i (1.13)	1739i (1.11)	1358i (1.16)	832i (1.26)
RB3-LYP/6-311 + G(3df,2pd)	1725i (1.08)	1656i (1.12)	1770i (1.09)	1392i (1.13)	926i (1.13)
UB3-LYP/6-31G(d)	1657i (1.13)	1592i (1.17)	1673i (1.15)	1311i (1.20)	737i (1.42)
UB3-LYP/6-31G(2df,p)	1638i (1.14)	1568i (1.19)	1649i (1.17)	1286i (1.22)	674i (1.55)
UB3-LYP/6-311G(d,p)	1638i (1.14)	1579i (1.18)	1659i (1.16)	1306i (1.20)	790i (1.32)
UB3-LYP/6-311 + G(3df,2pd)	1657i (1.13)	1602i (1.16)	1697i (1.13)	1352i (1.16)	891i (1.17)
UMPW1K/6-31 + G(d,p)	1773i (1.05)	1739i (1.07)	1824i (1.05)	1456i (1.08)	827i (1.26)
UMP2/6-31G(d)	2134i (0.87)	2061i (0.90)	2197i (0.88)	1849i (0.85)	1358i (0.77)
UMP2/6-311G(d,p)	1959i (0.95)	1865i (1.00)	2050i (0.94)	1600i (0.98)	1221i (0.86)
UMP2/6-311 + G(3df,2pd)	1941i (0.96)	1829i (1.02)	2038i (0.94)	1569i (1.00)	1185i (0.88)
UCCSD(T)/6-31G(d)	2029i (0.92)	2010i (0.93)	2073i (0.93)	1765i (0.89)	1126i (0.93)
UCCSD(T)/6-311G(d,p)	1866i (1)	1859i (1)	1924i (1)	1574i (1)	1046i (1)

^aThe numbers in parentheses are the scale factors required to convert the imaginary frequency into the corresponding CCSD(T)/6-311G(d,p) value.

Table 5. Effect of level of theory on the Eckart tunnelling corrections (Q_t)^a for the hydrogen abstraction reactions $\bullet\text{CH}_2\text{X} + \text{CH}_3\text{Y} \rightarrow \text{CH}_3\text{X} + \bullet\text{CH}_2\text{Y}$.

Level of theory	Reaction: $\bullet\text{CH}_2\text{X} + \text{CH}_3\text{Y} \rightarrow \text{CH}_3\text{X} + \bullet\text{CH}_2\text{Y}$				
	X,Y = H,H	X,Y = H,CN	X,Y = H,F	X,Y = H,Li	X,Y = F,Li
	(1)	(2)	(3)	(4)	(5)
RHF/6-31G(d)	7.8E + 05	3.2E + 04	4.0E + 05	1.2E + 04	7.2
RHF/6-311G(d,p)	5.9E + 05	2.8E + 04	3.3E + 05	1.0E + 04	7.1
RHF/6-311 + G(3df,2pd)	5.6E + 05	2.7E + 04	3.3E + 05	9.8E + 03	7.1
UHF/6-31G(d)	1.2E + 04	2.6E + 03	9.5E + 03	8.0E + 02	5.0
UHF/6-311G(d,p)	1.0E + 04	2.2E + 03	7.9E + 03	6.9E + 02	4.9
UHF/6-311 + G(3df,2pd)	1.1E + 04	2.3E + 03	8.4E + 03	7.3E + 02	5.0
RB3-LYP/6-31G(d)	9.7E + 01	4.1E + 01	9.4E + 01	1.0E + 01	1.8
RB3-LYP/6-311G(d,p)	8.2E + 01	3.7E + 01	8.3E + 01	9.8E + 00	1.8
RB3-LYP/6-311 + G(3df,2pd)	8.8E + 01	4.0E + 01	9.9E + 01	1.1E + 01	2.0
UB3-LYP/6-31G(d)	5.9E + 01	3.0E + 01	5.7E + 01	8.2E + 00	1.7
UB3-LYP/6-31G(2df,p)	5.2E + 01	2.7E + 01	5.0E + 01	7.5E + 00	1.5
UB3-LYP/6-311G(d,p)	5.2E + 01	2.8E + 01	5.3E + 01	8.1E + 00	1.8
UB3-LYP/6-311 + G(3df,2pd)	5.9E + 01	3.1E + 01	6.5E + 01	9.6E + 00	2.0
UMPW1K/6-31 + G(d,p)	1.2E + 02	6.0E + 01	1.4E + 02	1.4E + 01	1.8
UMP2/6-31G(d)	1.1E + 03	2.9E + 02	1.1E + 03	7.8E + 01	3.1
UMP2/6-311G(d,p)	3.7E + 02	1.1E + 02	5.0E + 02	2.6E + 01	2.8
UMP2/6-311 + G(3df,2pd)	3.3E + 02	9.3E + 01	4.7E + 02	2.3E + 01	2.7
UCCSD(T)/6-31G(d)	5.8E + 02	2.2E + 02	5.7E + 02	5.4E + 01	2.5
UCCSD(T)/6-311G(d,p)	2.1E + 02	1.1E + 02	2.4E + 02	2.4E + 01	2.3

^a Eckart tunnelling corrections (298 K) were calculated using the imaginary frequency at the specific level of theory listed and the CBS-RAD values for the forward and reverse barriers.

factors have been extremely successful. Of all of the levels of theory examined in the present work, the B3-LYP/6-311 + G(3df,2pd) calculations show the least variation (with scale factors ranging from 1.13–1.17), and thus represent the most promising prospect for the development of a general scale factor—although calculations on a larger test set (involving other types of hydrogen-atom abstraction reactions) would be needed to confirm this.

Given the large errors observed in the low-level imaginary frequencies, it is of interest to examine the effect of these errors on the calculated tunnelling coefficients. To this end, the imaginary frequencies in table 4 were used (in conjunction with the CBS-RAD reaction barriers and enthalpies in table 3) to obtain simple Eckart tunnelling coefficients (at 298 K) for the various hydrogen abstraction reactions (see table 5). It can be seen that the use of HF frequencies usually leads to errors of several orders of magnitude in the resulting tunnelling coefficients. Hence, HF methods (which are widely used as a low-cost procedure for the study of large systems) should be avoided for this type of calculation.

Fortunately, however, the errors are much smaller for the DFT and MP2 methods, with the best methods (MPW1K/6-31 + G(d,p), B3-LYP/6-311 + G(3df,2pd), MP2/6-311G(d,p) and MP2/6-311 + G(3df,2pd)) agreeing with our best estimates (CCSD(T)/6-311G(d,p)) to

within a factor of 2–3. The failure of the majority of the DFT methods to describe well the imaginary frequency for the strongly polar reaction (5) does not translate into large errors in the tunnelling coefficients because tunnelling is not very significant in this reaction. This is probably no coincidence as it is the strong polar interactions (which we suggested might be responsible for the poor imaginary frequencies) that lower the reaction barrier, leading in turn to a flatter reaction path and thereby reducing the contribution of tunnelling. Hence, DFT methods may perhaps be expected generally to perform reasonably well for tunnelling coefficients, despite their tendency to over-flatten the reaction path for strongly polar reactions. Furthermore, as noted above, the B3-LYP/6-311 + G(3df,2pd) imaginary frequencies can be improved through the use of a scale factor (of 1.15). When this is done, the errors in the calculated tunnelling coefficients diminish to factors of just 0.9–1.3. However, as noted above, this scale factor is based on a very small test set of reactions and calculations on a much wider test are required to investigate its general applicability.

In summary, the imaginary frequency is very sensitive to the level of theory used. The widely used HF methods severely overestimate the imaginary frequency, leading to the corresponding tunnelling coefficients being overestimated by several orders of magnitude.

These methods should therefore be avoided for tunnelling calculations. The moderate- and large-basis MP2 methods also overestimate the imaginary frequency, although the errors are much smaller (up to 17%) and translate into errors of only a factor of 2–3 in the tunnelling coefficient. Unfortunately, these methods are still too computationally expensive for the study of larger systems (such as those commonly encountered in biological reactions). A more cost-effective alternative is offered by the hybrid DFT methods, MPW1K/6-31+G(d,p) and B3-LYP/6-311+G(3df,2pd), which also produce errors in the tunnelling coefficients of a factor of 2–3, although in the opposite direction to the MP2 methods. Most of the DFT methods show significantly larger errors in the imaginary frequencies for the strongly polar reaction (5) but, since tunnelling is not very significant in this reaction, the tunnelling coefficients are not further compromised. Hence, DFT methods may be used to obtain reasonable (within a factor of 2–3) and cost-effective estimates of the tunnelling coefficients and this encouraging result complements earlier recommendations for the use of hybrid density functional methods, such as MPW1K [23], as low-cost procedures for the prediction of transition structure geometries and barrier heights [25].

4.2. IRCmax imaginary frequencies

Given the increasing numbers of computational studies of hydrogen abstraction in biologically relevant systems, which are often too large for high-level calculations, it is desirable to develop reliable low-cost procedures for calculating the electronic structure information needed for the estimation of tunnelling coefficients. As seen above, hybrid DFT methods such as MPW1K and B3-LYP can be used to obtain reasonable estimates of the tunnelling coefficients (to within a factor of 2–3). Nevertheless, it is of interest to develop cost-effective procedures in which the errors in the low-level imaginary frequencies can be yet further reduced. To this end, we investigated the performance of the IRCmax procedure for calculating imaginary frequencies.

As shown in §3, it is possible to estimate the imaginary frequency by fitting a parabolic function to points on the MEP of a reaction, in the vicinity of the transition structure. IRCmax imaginary frequencies can therefore also be estimated by fitting a parabolic function but in this case to high-level single-point energies calculated along the low-level MEP. As a justification of this approach, we could view the high-level mass-weighted force constant matrix, $\mathbf{F}_\mu^{(H)}$, at some point on the low-level MEP as a perturbation of the

corresponding low-level force constant matrix, $\mathbf{F}_\mu^{(L)}$:

$$\mathbf{F}_\mu^{(H)} = \mathbf{F}_\mu^{(L)} + \delta\mathbf{F}. \quad (10)$$

Let $\mathbf{v}^{(L)}$ denote the eigenvector of $\mathbf{F}_\mu^{(L)}$ corresponding to the imaginary frequency and $\lambda^{(L)}$ the associated eigenvalue of $\mathbf{F}_\mu^{(L)}$. Then, in first-order perturbation theory, the corresponding high-level eigenvalue, $\lambda^{(H)}$, is approximated by

$$\lambda^{(H)} \approx \lambda^{(L)} + \mathbf{v}^{(L)T} \delta\mathbf{F} \mathbf{v}^{(L)}, \quad (11)$$

where T denotes the transposed vector. Thus,

$$\lambda^{(H)} \approx \mathbf{v}^{(L)T} \mathbf{F}_\mu^{(H)} \mathbf{v}^{(L)}. \quad (12)$$

Evaluating the right-hand side of equation (12) by finite differences means evaluating the second-order difference of the high-level energy on the low-level MEP. Hence, if the low-level reaction path is qualitatively correct, we could anticipate that the high-level imaginary frequency would be reasonably approximated by finite differences on the low-level path.

IRCmax imaginary frequencies, calculated using high-level CCSD(T)/6-311G(d,p) single-point energies, along various low-level MEPs for the hydrogen abstraction reactions (1)–(5), are shown in table 6. In calculating the IRCmax imaginary frequencies, the optimal conditions for estimating an imaginary frequency from a MEP (as determined above) were used. Included in table 6 are the corresponding exact imaginary frequencies at both the high level used for the IRCmax single points, and the various low levels used to calculate the MEPs. From these results it can be seen that IRCmax can offer spectacular improvements to the estimated imaginary frequencies, although its success is dependent on the level of theory used to calculate the low-level MEP. For HF/6-31G(d), the IRCmax technique enables the high-level CCSD(T)/6-311G(d,p) imaginary frequencies to be reproduced to within 4%. This translates to predicting tunnelling coefficients to within a factor of 0.9–1.6. In contrast, the exact imaginary frequencies at this low level of theory show errors of 36–100%, and corresponding errors in the predicted tunnelling coefficients of up to two orders of magnitude. In the case of the B3-LYP/6-31G(2df,p) MEP, the errors in the IRCmax imaginary frequencies for reactions (1)–(4) also diminish to 4%, compared with the errors of 12–18% for the corresponding exact values at the low level of theory. However, while the error for the polar reaction (5) does decrease from 36% to 24%, it remains quite large. In the case of the MP2/6-311G(d,p) MEP, only reactions (1) and (2) were investigated in detail because it was found

Table 6. Comparison of IRCmax and exact imaginary frequencies ($\omega^\ddagger$, cm^{-1}) for the hydrogen abstraction reactions $\bullet\text{CH}_2\text{X} + \text{CH}_3\text{Y} \rightarrow \text{CH}_3\text{X} + \bullet\text{CH}_2\text{Y}$.

Level of theory	Reaction: $\bullet\text{CH}_2\text{X} + \text{CH}_3\text{Y} \rightarrow \text{CH}_3\text{X} + \bullet\text{CH}_2\text{Y}$				
	X,Y = H,H	X,Y = H,CN	X,Y = H,F	X,Y = H,Li	X,Y = F,Li
	(1)	(2)	(3)	(4)	(5)
<i>Exact $\omega^\ddagger$^a</i>					
UHF/6-31G(d)	2558i	2578i	2611i	2421i	2096i
UB3-LYP/6-31G(2df,p)	1638i	1568i	1649i	1286i	674i
UMP2/6-311G(d,p)	1959i	1865i	2050i	1600i	1221i
UCCSD(T)/6-311G(d,p)	1866i	1859i	1924i	1574i	1046i
<i>IRCmax $\omega^\ddagger$ at CCSD(T)/6-311G(d,p)</i>					
//UHF/6-31G(d)	1939i	1850i	1968i	1597i	1010i
//UB3-LYP/6-31G(2df,p)	1827i	1875i	1939i	1505i	792i
//UMP2/6-311G(d,p)	1761i	1685i	—	—	—

^aThis is the directly calculated imaginary frequency at the particular level of theory.

that the error actually increases on applying the IRCmax technique.

In summary, the IRCmax technique can be used to obtain excellent approximations to the high-level CCSD(T)/6-311G(d,p) imaginary frequencies, at a fraction of the cost of a full frequency calculation at that level. However, its success is (not surprisingly) dependent on the level of theory used to calculate the low-level MEP. For the present hydrogen abstraction reactions, it is found that the low-cost HF/6-31G(d) method actually provides the best surface (of the three tested) for applying the IRCmax technique and enables the high-level CCSD(T)/6-311G(d,p) imaginary frequencies to be reproduced to within 4%.

5. Conclusions

In the present work we have shown that the calculated imaginary frequency for hydrogen abstraction reactions is very sensitive to the level of theory used, and this in turn can lead to significant errors in the calculated tunnelling coefficients for these reactions. In particular, the widely used HF methods severely overestimate the imaginary frequencies, leading to errors in the calculated tunnelling coefficients of several orders of magnitude. These methods should therefore be avoided for tunnelling calculations. The hybrid DFT methods are found to significantly underestimate the imaginary frequency, especially for the polar reaction $\bullet\text{CH}_2\text{F} + \text{CH}_3\text{Li} \rightarrow \text{CH}_3\text{F} + \bullet\text{CH}_2\text{Li}$. However, for the best methods (MPW1K/6-31 + G(d,p) and B3LYP/6-311 + G(3df, 2pd)), the errors in the imaginary frequencies result in an underestimation of the tunnelling coefficient by only a factor of 2–3. The MP2 methods are found to overestimate the tunnelling coefficients by a similar amount.

The scale factors required to correct the errors in the imaginary frequencies are found generally to vary significantly among the reactions examined. This is in stark contrast to the widespread success of scale factors in correcting errors in the calculated normal-mode vibrations of molecules. However, one exception to this is the B3-LYP/6-311 + G(3df,2pd) level where the use of a scale factor of 1.15 for the imaginary frequency looks promising, although further calculations on a much wider test set are required to examine the generality of this result.

An alternative strategy for obtaining improved estimates of the imaginary frequency is via the IRCmax procedure developed in the present work. In this technique, the imaginary frequency is obtained by fitting a parabolic function to high-level single-point energies along a low-level minimum energy path (MEP). We find that the IRCmax imaginary frequencies estimated using CCSD(T)/6-311G(d,p) single points along a low-level HF/6-31G(d) MEP provide an excellent approximation (to within 4%) to the exact imaginary frequencies obtained from a full frequency calculation at the CCSD(T)/6-311G(d,p) level, at a fraction of the computational cost. Hence the use of such a technique could greatly improve the accuracy of tunnelling calculations for large systems where high-level frequency calculations are not feasible.

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